Stage 2 Report: Data Pipeline and First Baseline

Image Super-Resolution with Deep Generative Models · Computer Vision 2026 · 13 Oct 2026 Team: Julia Kurnaeva, Alena Petrenko, Amaliya Kharisova, Dmitrii Kuznetsov

Repository: https://github.com/Ephy01/Image-Super-Resolution-with-Deep-Generative-Models

Workflow changes: We mention this to prevent any misunderstanding. We decided that each week only one of us works on the project. At the end of the week, this team member explains their work to the others.

1. Data loading and preprocessing

The pipeline is two scripts. `download_data.py` fetches the raw data; `prepare_data.py` builds everything the models and metrics read.

Content	Source and preprocessing	Test	Dev
Objects & scenes	DIV2K validation 0801–0900; central 512×512 crop at native resolution	80	20
Faces	120 FFHQ images sampled with a fixed seed from 00000–00999; resized 1024 → 512	100	20
Text	Rendered by us: 3 DejaVu fonts × 4 font sizes (16, 24, 32, 48 px) × dictionary words or random codes; the exact text is stored	96	24

- **Low-resolution inputs.** Each 512×512 image is reduced ×4 with bicubic interpolation to 128×128 and JPEG-compressed at quality none, 50, 30 and 10 (4:2:0). The result is stored as PNG, so the only compression is the one applied on purpose. 340 images × 4 levels = 1,360 inputs.
- **Split.** No training split, because nothing is trained. Dev (64 images) is for debugging and choosing settings. Test (276 images) is kept for the final numbers and has not been run. The split is by source image; dev and test text are rendered with different seeds.
- **Checks.** The script stops if a source image or a text appears in both splits, and writes the image counts to `data_report.json`. All random choices use a fixed seed.

2. Baseline models and configuration

- **Bicubic interpolation** (Pillow): no learning; the reference every model must beat.
- **Real-ESRNet x4plus**: the first pretrained baseline. RRDBNet with 23 residual-in-residual dense blocks, 64 feature channels and 32 growth channels: 16.7 M parameters. Official pretrained weights, loaded into our own definition of the network. Input 128×128, output 512×512, batch size 1, float32, no tiling.

3. Training setup

No model is trained or fine-tuned. Both baselines run as they are on the dev split. Settings are chosen on dev only, and the test split is run once at the end.

4. Initial quantitative results (dev split, 64 images)

PSNR and SSIM on the Y channel with a 4-pixel border crop; LPIPS with AlexNet. Mean over images.

Content	Method	PSNR ↑, no JPEG	SSIM ↑	LPIPS ↓	PSNR ↑, JPEG 10	SSIM ↑	LPIPS ↓
Objects	Bicubic	25.62	0.689	0.488	22.77	0.543	0.767
Objects	Real-ESRNet	**26.31**	**0.737**	**0.376**	**22.97**	**0.582**	**0.613**
Faces	Bicubic	33.40	0.879	0.290	28.25	0.769	0.647
Faces	Real-ESRNet	**33.76**	**0.894**	**0.238**	**28.72**	**0.802**	**0.456**
Text	Bicubic	16.46	0.707	0.436	15.12	0.581	0.639
Text	Real-ESRNet	**19.31**	**0.838**	**0.186**	**15.67**	**0.690**	**0.408**

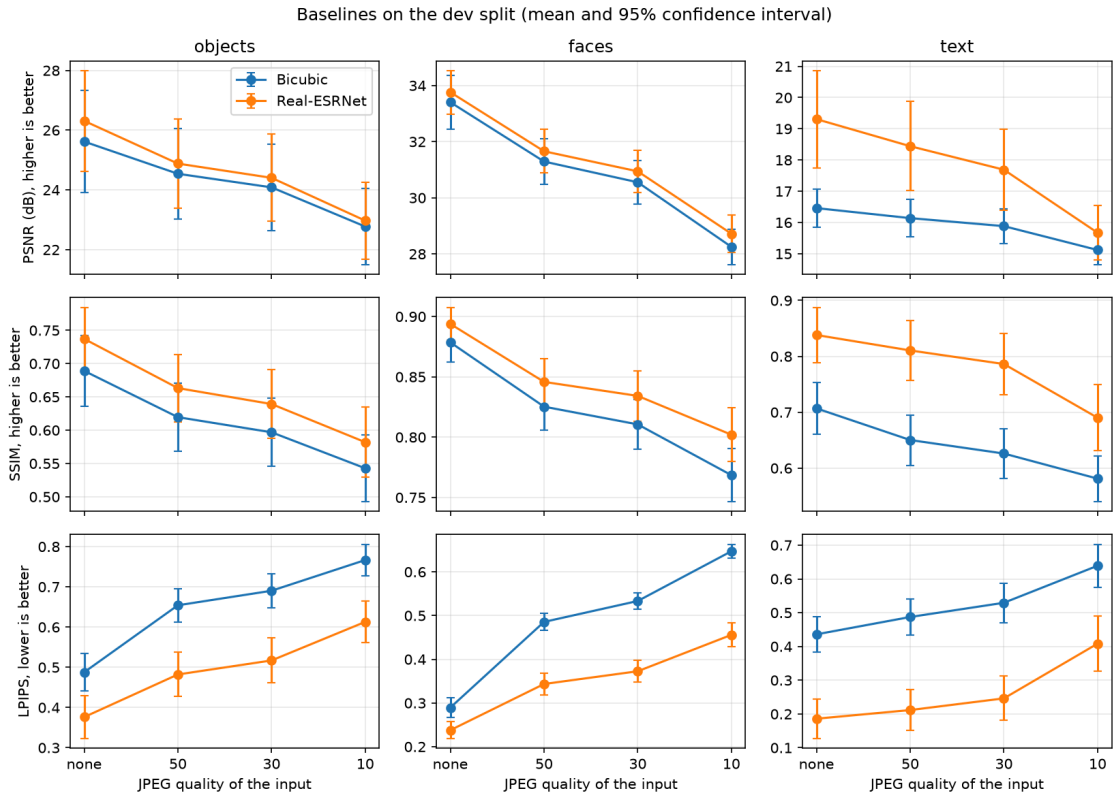
Paired difference Real-ESRNet − bicubic on the same images, no JPEG, with 95% bootstrap interval: PSNR +0.69 ± 0.32 dB (objects), +0.36 ± 0.26 dB (faces), +2.85 ± 0.95 dB (text); LPIPS −0.11, −0.05, −0.25. Runtime on an Apple M4 GPU: bicubic 0.001 s, Real-ESRNet 0.28 s per image.

5. Problems and observations

- **The PSNR gain on photos is small, the LPIPS gain is large.** Real-ESRNet adds only 0.4–0.7 dB on objects and faces but lowers LPIPS by 18–23%. Sharper edges matter more to LPIPS than to PSNR.
- **On text, the gain depends on letter size.** With letters 12 px tall in the input, Real-ESRNet gains 6.1 dB; at 4 px it gains 0.2 dB. At 6 px its letters are sharp but visibly reshaped, which PSNR and LPIPS cannot judge. This is why an OCR error rate is planned.
- **Heavy compression removes most of the PSNR advantage.** At JPEG 10 the gain falls to 0.2 dB on objects and 0.6 dB on text.
- **Dev intervals are wide.** With 20 images per content type, the mean PSNR on objects is ± 1.7 dB. Method comparisons therefore use paired differences, which are much tighter.
- **FFHQ download.** The official release is one 89 GB folder. We download single images from a mirror that stores one file per image.

6. Figures

Metrics against the JPEG quality of the input (mean and 95% confidence interval, dev split):



Central 256×256 crops, input without JPEG. The text row has letters 6 px tall in the input:

